

MANOGNA VADLAMUDI

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EDUCATION

Illinois Institute of Technology, Chicago, IL

MAY 2025

Master of Computer Science

Sridevi Women's Engineering College, Hyderabad, India (JNTUH)

JUL 2023

Bachelor of Technology

SKILLS

Programming Languages: JavaScript, C, Python, SQL, Java (basics), Object Oriented Programming.

Frameworks & Libraries: React JS, Angular, Flask, Node.js, Express.js, RESTful APIs, HTML, CSS, Material UI.

Machine Learning: Predictive analytics, NLP, Scikit, NumPy, SciPy, Pandas, NLTK, matplotlib, and seaborn.

Database Technologies: MySQL, MongoDB, Elasticsearch.

Cloud Services: Amazon Web Services (AWS), Google Cloud Services.

Tools: GIT, GitHub, VS Code, Android Studio, Jupyter lab, Postman, Microsoft suite.

PROJECTS

Smart Ride-Sharing System with AI-Powered Insights:

JAN 2025 – MAY 2025

- Developed a full-stack ride-sharing platform with Angular, Node.js, Express, and MongoDB, supporting role-based access (Rider, Driver, Admin) and integrated RESTful APIs.
- Integrated ML modules (Python) for ETA and fare predictions using the Strategy pattern, and implemented Singleton and Factory Method design patterns for core logic.
- Built an analytics dashboard with Chart.js and chatbot module; followed MVC and component-based architecture for scalability and maintainability.

Leveraging Machine Learning for Autism Spectrum Disorder (ASD) Detection:

JAN 2024 – APR 2024

- Achieved 96% ASD classification accuracy by training and tuning Logistic Regression, Decision Tree, and Random Forest with GridSearchCV and stratified 5-fold CV.
- Preprocessed data via median/mode imputation, Z-score/IQR outlier removal, and correlation-based feature pruning. Automated encoding and scaling in ColumnTransformer pipelines (one-hot/ordinal + StandardScaler/MinMax).

Path-Hole Detection and Alert System in Vehicles:

FEB 2023 – MAY 2023

- Built a system using sensors, Arduino, and algorithms to detect potholes and speed breakers, enhancing road safety.
- Utilized Arduino to calculate distances and trigger alerts upon detecting unsafe road conditions using predefined equations and thresholds.
- Supported ongoing research in pothole detection systems to prevent accidents and improve road infrastructure monitoring.

Supermarket Billing System Using Webcam:

AUG 2022 – DEC 2022

- Developed a computer vision-based automated billing system using webcam technology for real-time product detection.
- Replaced traditional barcode scanning with AI-powered product recognition, reducing checkout time, and improving accuracy.
- Integrated automated price calculation and billing to streamline retail operations and enhance customer experience.

INTERNSHIPS

Artificial Intelligence Internship - Internship Studio

JUN 2022

- Built a basic English-to-German translation tool using Python and Google Cloud Translation API with text preprocessing and batched API calls.
- Gained hands-on experience in Dockerizing the tool and integrating it as a simple CLI utility, exploring fundamentals of deployment and automation.

Amazon Web Services Internship - BrainOVision Pvt. Ltd.

APR 2022 – MAY 2022

- Gained hands-on familiarity with core AWS services—**launched** EC2 instances, configured S3 buckets, and set up basic IAM users, roles, and policies via the AWS Console and CLI.
- Explored foundational infrastructure-as-code and monitoring practices by reviewing simple CloudFormation templates and viewing resource metrics in CloudWatch.

HONORS AND ACHIEVEMENTS

- Member of the Illinois Tech Chapter of Upsilon Pi Epsilon
- Red Hat Enterprise Linux Technical Overview
- Career Skills TCS ion
- CCSK v4.1 Foundation Training Cloud Security Alliance
- STEP Course by HINDU Group with score of 9.1

APRIL 2025

FEB 2025

APR 2021

MAY 2022

JUL 2020